Pupillometry Remains Sensitive to SNR-based Manipulations Across Prolonged Experimental Sessions

update on an ongoing project (Oct. 2025)

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GAMMs Club
04.08.2026

University of Minnesota
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Who We Are

- **Brady Chisholm**: U of MN Psychology B.S. August 2025, post-bacc researcher in Auditory Perception and Cognition Lab
- **Juraj Mesik**: Research Scientist at U of MN, predominantly interested in cortical processing of speech in challenging listening backgrounds (using TRF methods). Interested in the possible interplay of listening effort & cortical tracking of various speech features
- Both new to GAMMs, but learning more since Spring 2025 :)
- Working towards manuscript draft on fatigue… (more to come)

If you were a reviewer, what's your verdict? ✅/❌

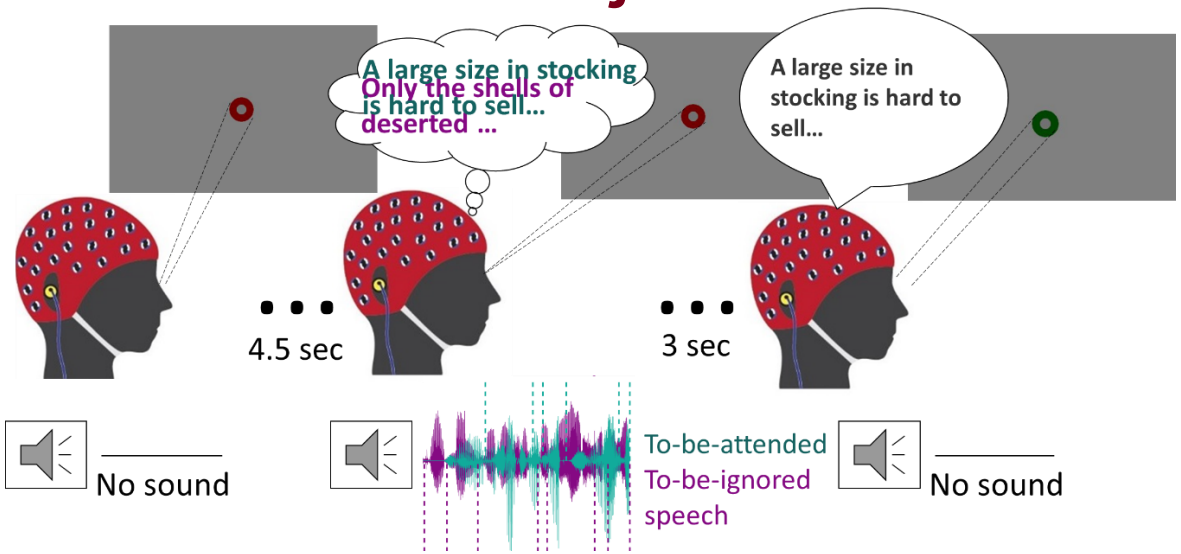
Research Question

- Motivation: Pupillometry is a mainstream measurement

Citations of “Pupillometry” / 100,000 articles on Pubmed (Sperr, 2016)	
2015 = 7.23	2025 = 17.58

- Unclear if fatigue interacts with effect of LE manipulation
  - Presence of fatigue (in some form) is acknowledged in much of the literature discussing pupillometry
- Goal: Analyze existing dataset to assess whether fatigue and LE interact (or not…)
  - Accomplish by assessing how time x block x SNR interact

Present Project



Paradigm: Diotic speech-in-speech recognition at four SNR levels (5, 0, -5, and -10 dB SNR) with 50 trials/condition, performed over 10 blocks of 20 trials. SNR varied on a trial-by-trial basis.

Stimuli: IEEE target sentences.

Distractor is simultaneous audiobook fixed at 65 dB SPL, and preceded target audio onset by 1 second each trial to facilitate streaming.

Pupillometry Data: Tobii Pro Spectrum tracking @ 300Hz

Load preprocessed pupillometry data from PUPILS Toolbox (Relaño-Iborra & Bækgaard, 2020) and then in R…

Data from PUPILS Toolbox → **Choose Best Eye for Stats** → (**Upsample if != 300Hz**) → **Bin Average to 10Hz** → **Prep GAMM dataframe** (factors, AR.start, etc.)

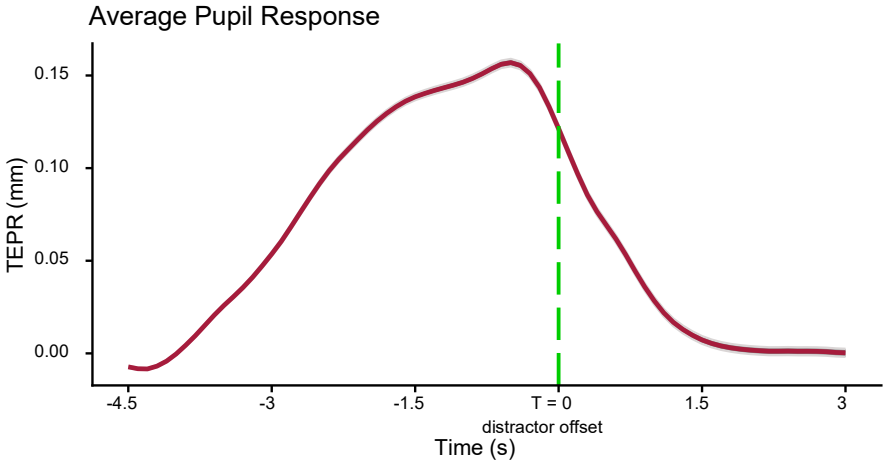
Pupil Data

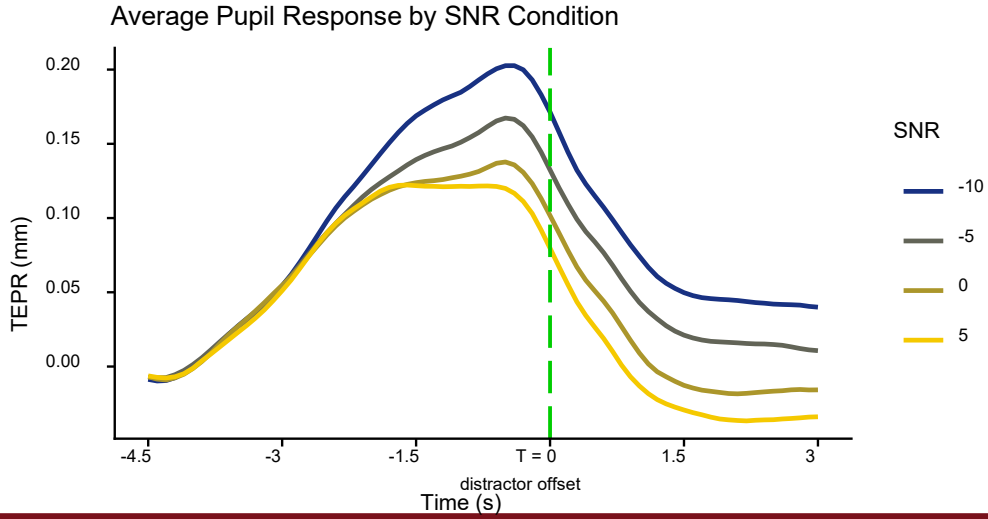
Part of a larger study focusing on LE involving EEG, self-report, etc.

Pupil *mean* = 0.063; *sd* = 0.198

N = 62, Age *mean*: 47.6; *sd*: 17.65

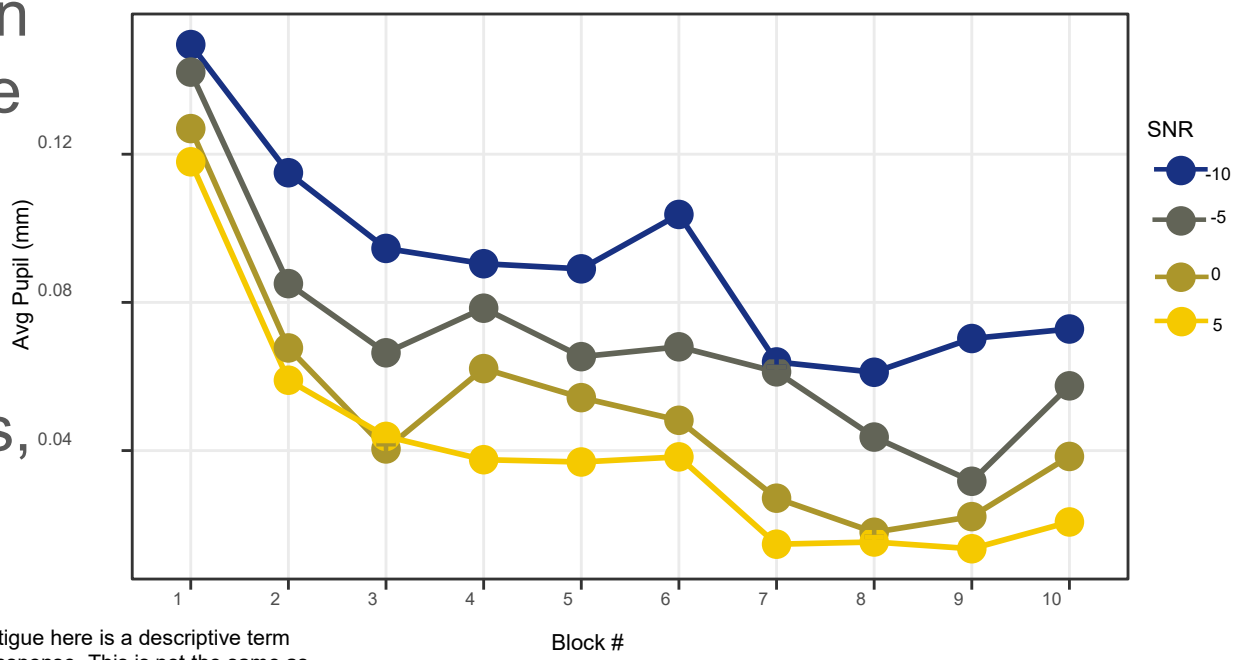
- Assert effect of SNR on pupil
- Offset aligned data
  - T = 0 is Distractor Offset
  - Target offset is -0.5s prior to T = 0





The Effect of Block (or…fatigue)

- Can observe fatigue in raw average response
- To most, weaker dilations over time is unwanted variance
- Effect of SNR persists, but does it interact with fatigue?



Note: *Pupillometric* fatigue here is a descriptive term for habituating pupil response. This is not the same as cognitive fatigue (in this presentation we are agnostic about the causes / contributing factors).

Previous Roadblocks

Issue

- Models failing to converge (endlessly running)
- Flat lines when plotting smooths using `itsadug`
- Exceeding memory capacity

Resolution

- Used offset aligned *vs* full time course
- Changed basis function
- Downsampled to 10Hz
- Bin averaging of data

How to Separate out Effects?

- Previous issue presented in Oct. 2025 meeting
  - Pursued binary predictors (such as Wieling et al., 2018; Johns et al., 2024; Smith et al., 2025)
- Believed with 10 blocks and 4 SNRs, would result in 9 (block main effects) + 3 (snr main effects) + 27 (block*snr interaction effects) = 39 binary predictors
- Also used '`s z`' smooths at the time

What We Actually Did

- Treated *only* SNR as an ordered factor variable
  - Excluded one level for reference treatment
  - Resulted in `isSNR_x` variables for each condition, modeled as parametrics

- Reference surfaces of `time` and `block`
- Differences surfaces of `block` by `isSNR_x`
- Interaction terms of `time x block` by `isSNR_x`
- `ti()` terms all the way…

Our Iterative Modeling Process

Model	Adj. R-sq	Dev. Explained
basicGAMM_noBlock_noAR1	*0.291*	*23.7%*
basicGAMM_noBlock_withAR1	*0.282*	*22.8%*
basicGAMM_noAR1	*0.317*	*25.7%*
basicGAMM_wAR1	*0.307*	*24.8%*
fullTensorGAMM	*0.309*	*25.0%*

Model of Interest

"Final" formula

- Use tp basis function
- 5.6 GB

Term	k value
Time	50
Block	10
Random Effects	40

```
tensorGAMM_acf_and_rho_FULLData <- bam(pupil ~
 # parametric main effects
   isSNR_m10 + isSNR_m5 + isSNR_0 +

 # reference surfaces
ti(time, k = seconds_k, bs = "tp") +
ti(block, k = block_k, bs = "tp") +

# reference interaction
ti(time, block, k = c(seconds_k, block_k), bs = "tp") +

# differences surfaces of time by isSNR_x
ti(time, by = isSNR_m10, k = seconds_k, bs = "tp") +
ti(time, by = isSNR_m5, k = seconds_k, bs = "tp") +
ti(time, by = isSNR_0, k = seconds_k, bs = "tp") +

# difference surface of block by isSNR_x
ti(block, by = isSNR_m10, k = block_k, bs = "tp") +
ti(block, by = isSNR_m5,  k = block_k, bs = "tp") +
ti(block, by = isSNR_0,   k = block_k, bs = "tp") +

# --- INTERACTIONS ---
# discern time * block differences by SNR_x compared to reference of SNR 5
ti(time, block, by = isSNR_m10, k = c(seconds_k, block_k), bs = "tp") +
ti(time, block, by = isSNR_m5,  k = c(seconds_k, block_k), bs = "tp") +
ti(time, block, by = isSNR_0,   k = c(seconds_k, block_k), bs = "tp") +
# --- RNDM EFFECTS ---
s(time, subject, by = isSNR_m10, bs = "fs", m = 1, k = kRandom) +
s(time, subject, by = isSNR_m5, bs = "fs", m = 1, k = kRandom) +
s(time, subject, by = isSNR_0, bs = "fs", m = 1, k = kRandom) +
s(time, subject, bs = "fs", m = 1, k = kRandom),
```

Results

What to plot?

What matters?

- Start with summary
- Identify significant terms
  (what's wiggly?)
- Investigate further via plotting

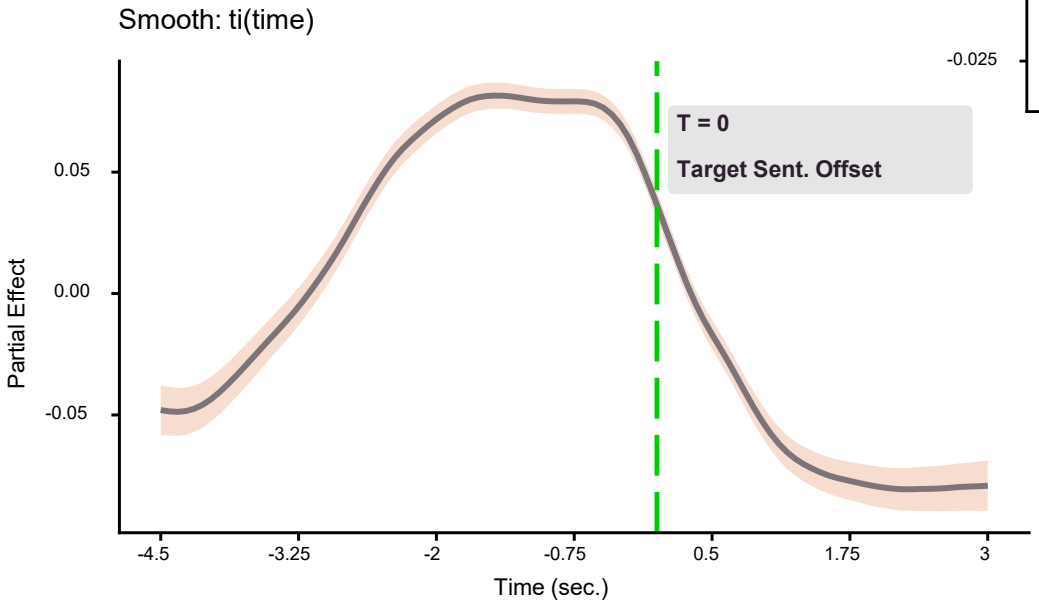
Tensor Interaction Model Summary Statistics

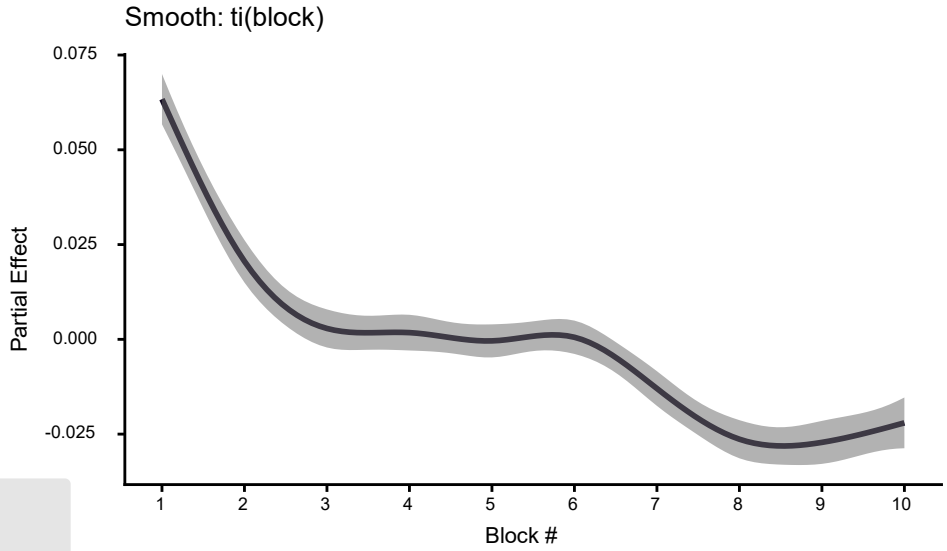
Term				
Smooth terms	*EDF*	*Ref.df*	*F*	*p*
time	34.49	36.60	47.72	< .001
block	7.41	8.39	46.36	< .001
block, time	216.10	282.20	4.08	< .001
time × isSNR_m10TRUE	18.57	22.77	16.18	< .001
time × isSNR_m5TRUE	14.79	18.94	9.42	< .001
time × isSNR_0TRUE	10.94	14.11	3.27	< .001
block × isSNR_m10TRUE	1.02	1.04	0.76	.390
block × isSNR_m5TRUE	1.01	1.02	0.00	.997
block × isSNR_0TRUE	1.03	1.06	0.07	.826
block, time × isSNR_m10TRUE	66.91	108.59	0.61	.999
block, time × isSNR_m5TRUE	87.69	138.86	0.72	.994
block, time × isSNR_0TRUE	42.63	63.15	1.06	.345
Random smooths	*EDF*	*Ref.df*	*F*	*p*
time, subject × isSNR_m10TRUE	1215.56	2480.00	1.05	< .001
time, subject × isSNR_m5TRUE	779.74	2480.00	0.47	< .001
time, subject × isSNR_0TRUE	796.57	2479.00	0.49	< .001
time, subject	2158.47	2480.00	9.26	< .001

Adjusted R-squared = 0.309 | Deviance Explained = 25.0% | fREML = 282911.95

Results (cont.)

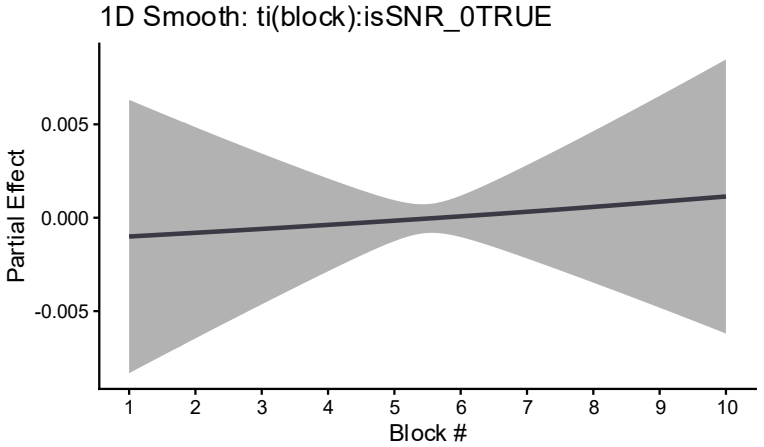
First inspect significant smooths of block and time

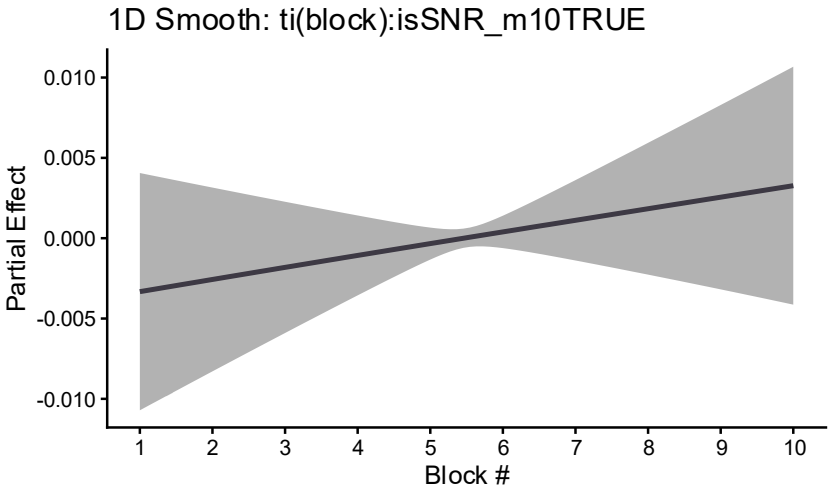


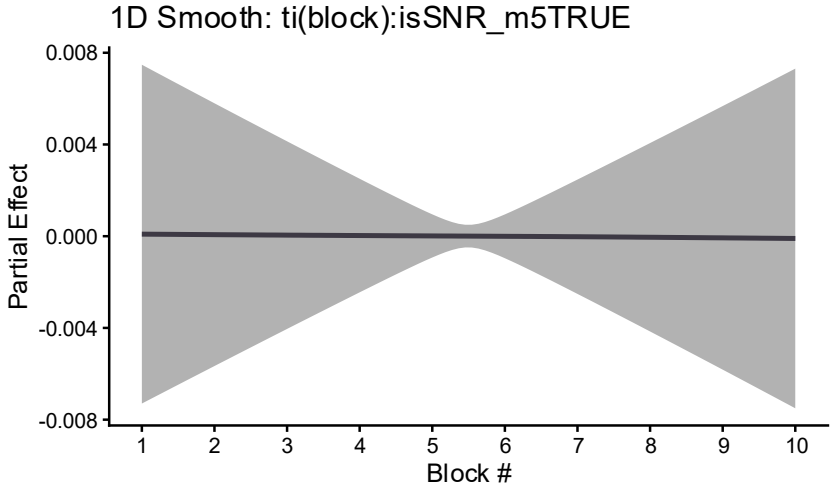


Results (cont.)

Also significant smooths of block x SNR, but unsure how to interpret… (and bowtie CIs!)



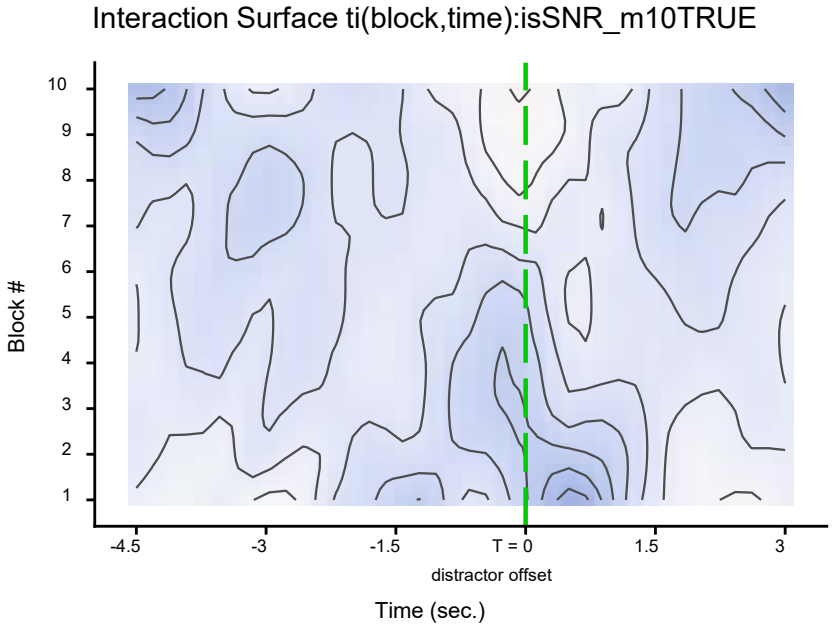


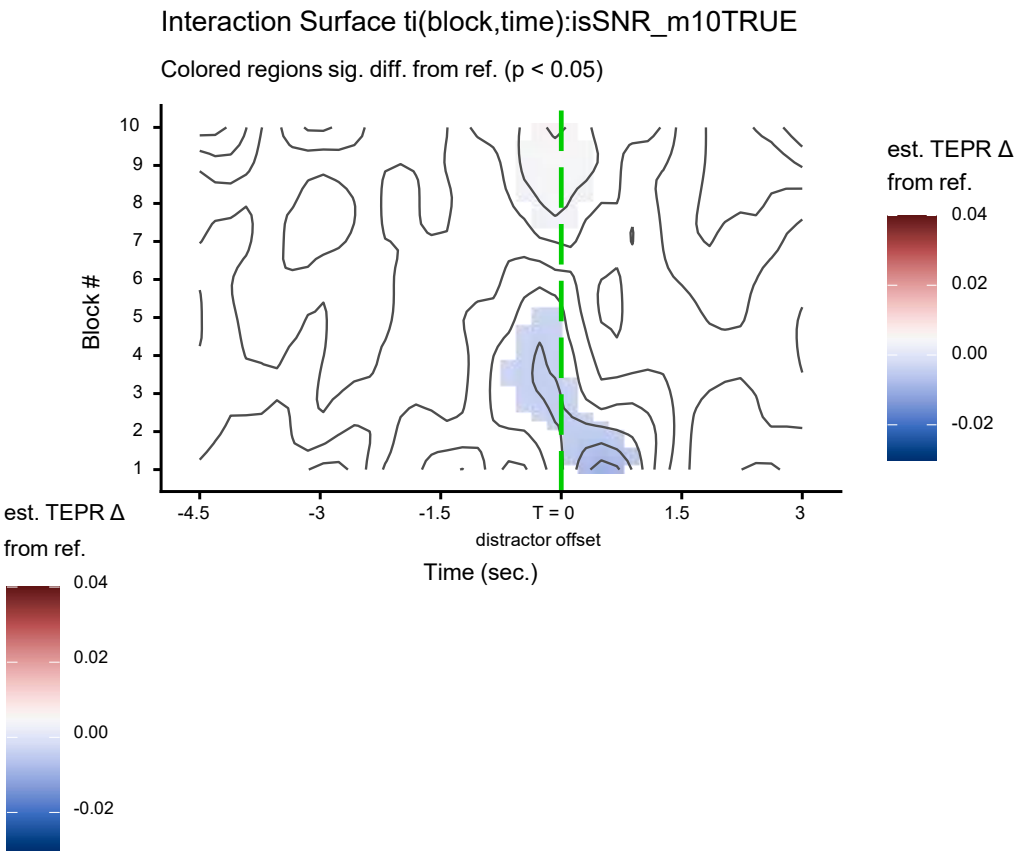


Results SNR -10

What does the interaction surface look like?

(with and w/o sig. test)

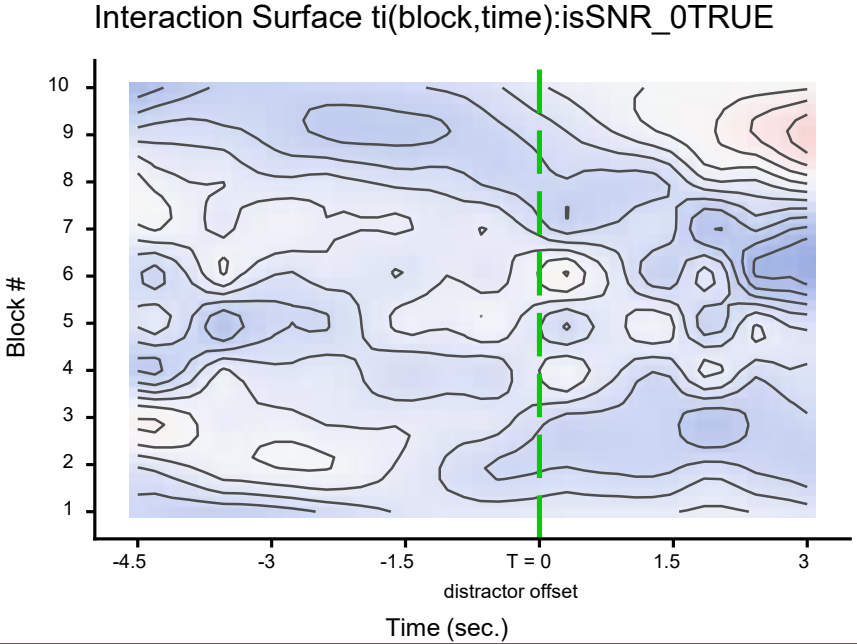


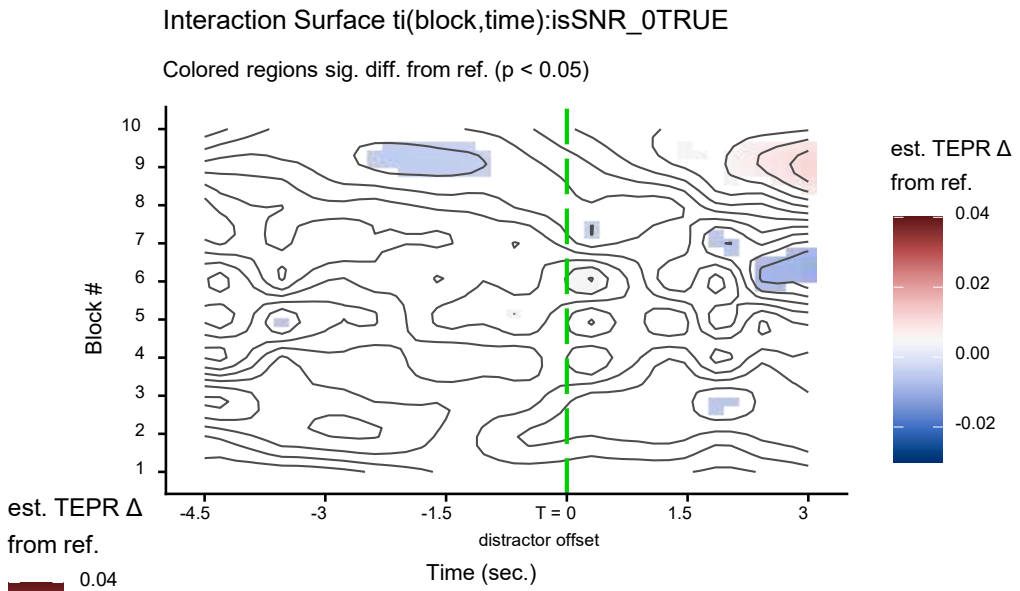


Results SNR 0

What does the interaction surface look like?

(with and w/o sig. test)



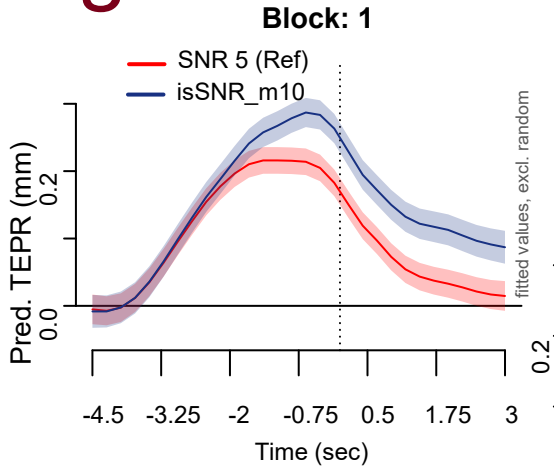


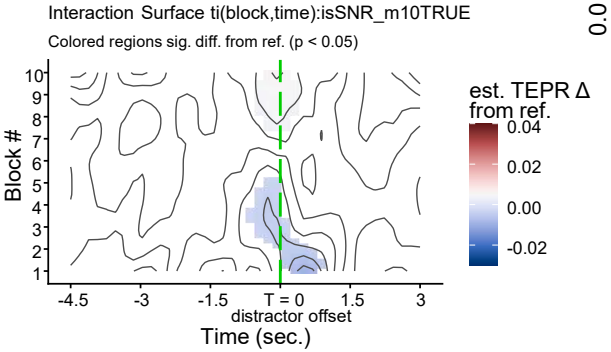
Comparing SNR over Block

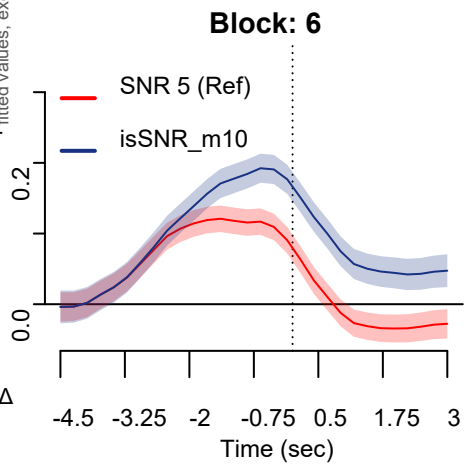
Observed areas of significant difference are nearly nonexistent.

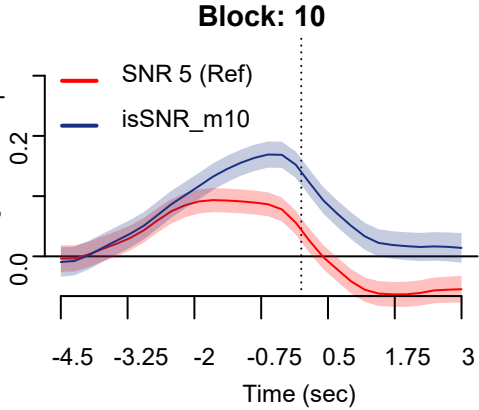
But what does the difference look like?

(itsadug plots)



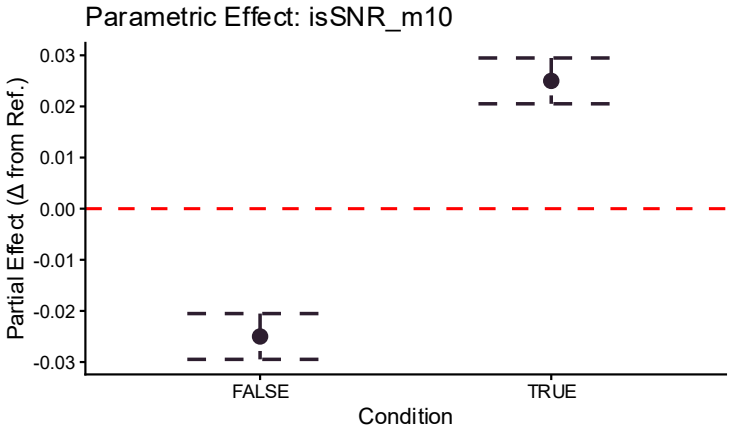


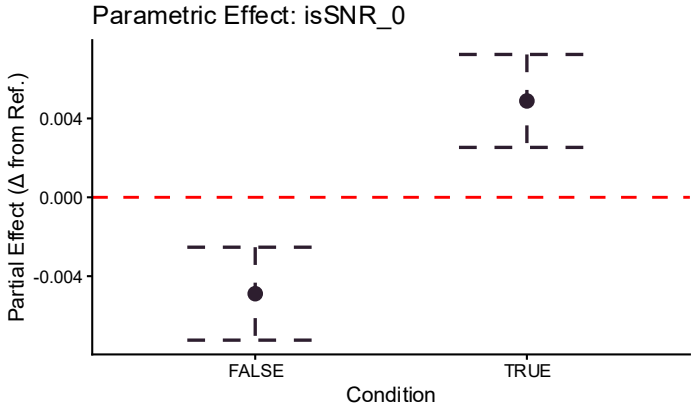


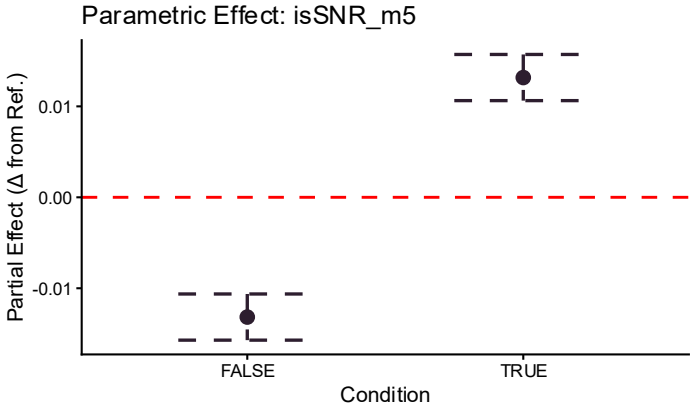


Results (cont.)

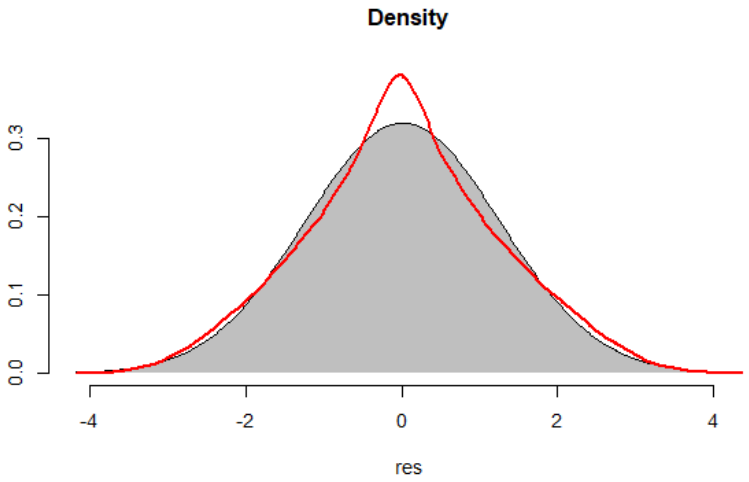
Parametrics

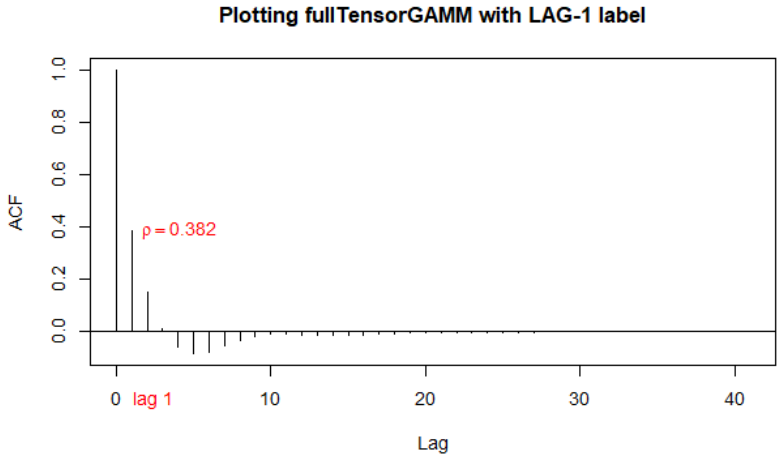


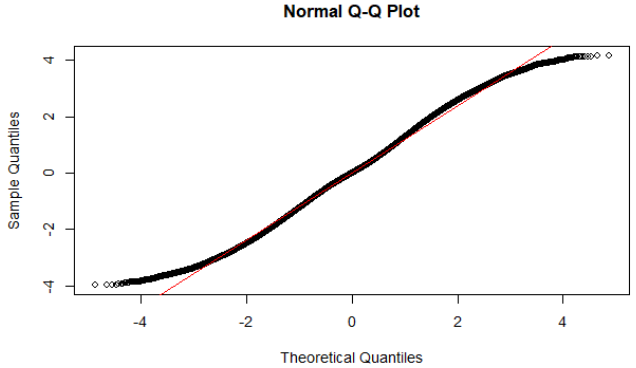




Critiquing our Model







Summary

- Fatigue does not interact with ***this*** manipulation of LE
- Uncertain how fatigue may behave in other paradigms

- Did model specifications, interpretation, figures make sense? What would help to see?

- If you were a reviewer…

Decision?
Accept ✅
Accept with revisions 📝
Rejected ❌

End slide



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Releveled Models

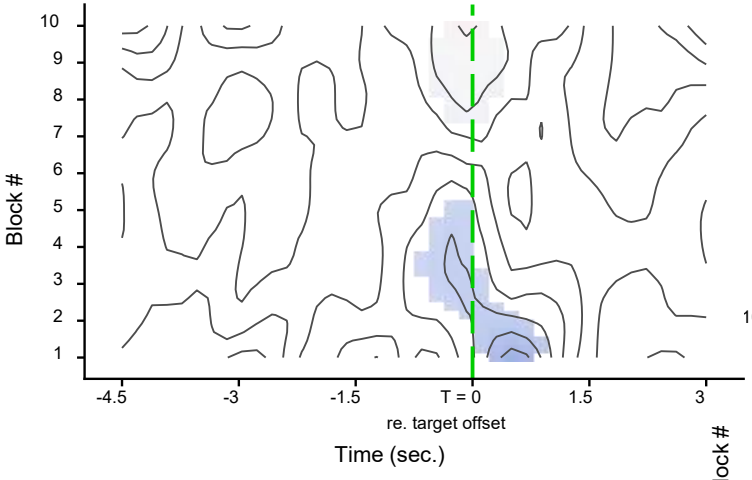
- Ran after plotting full model, wanted to quell any uncertainty about choice of reference condition
- Releveled to other SNRs, ran with identical formula

Model	AIC	Dev. Expl
releveledMod_ref0		
releveledMod_refM5		
releveledMod_refM10		
fullTensorGAMM	-3377403	

Results

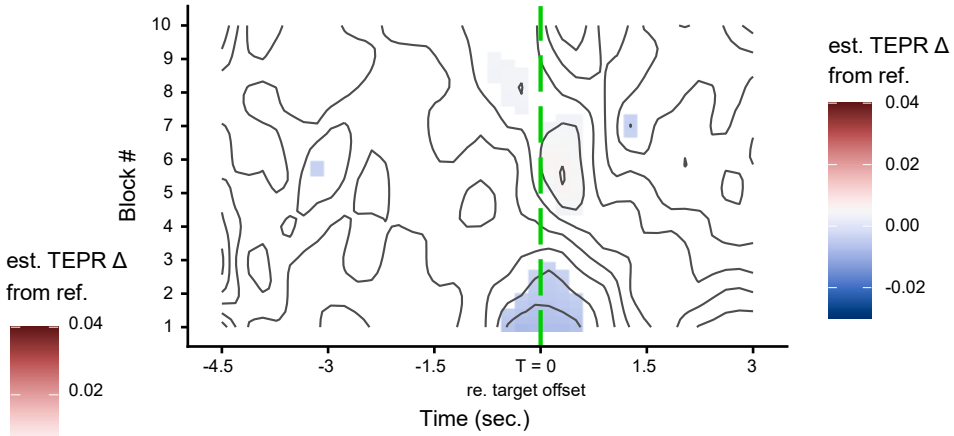
2D Surface: ti(block,time):isSNR_m10TRUE

Colored regions sig. diff. from ref. (p < 0.05)



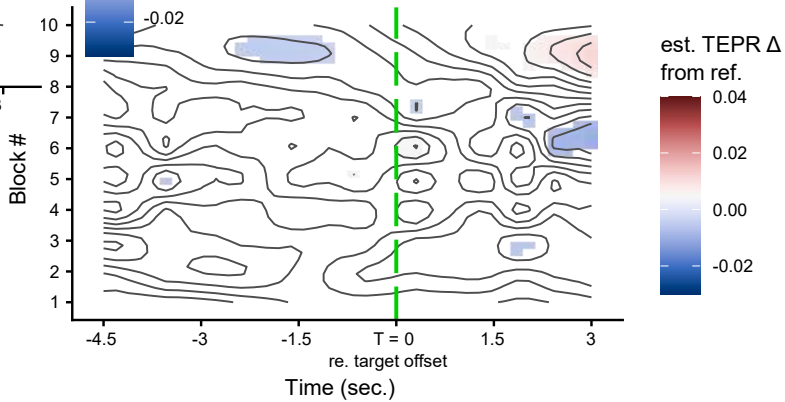
2D Surface: ti(block,time):isSNR_m5TRUE

Colored regions sig. diff. from ref. (p < 0.05)



2D Surface: ti(block,time):isSNR_0TRUE

Colored regions sig. diff. from ref. (p < 0.05)

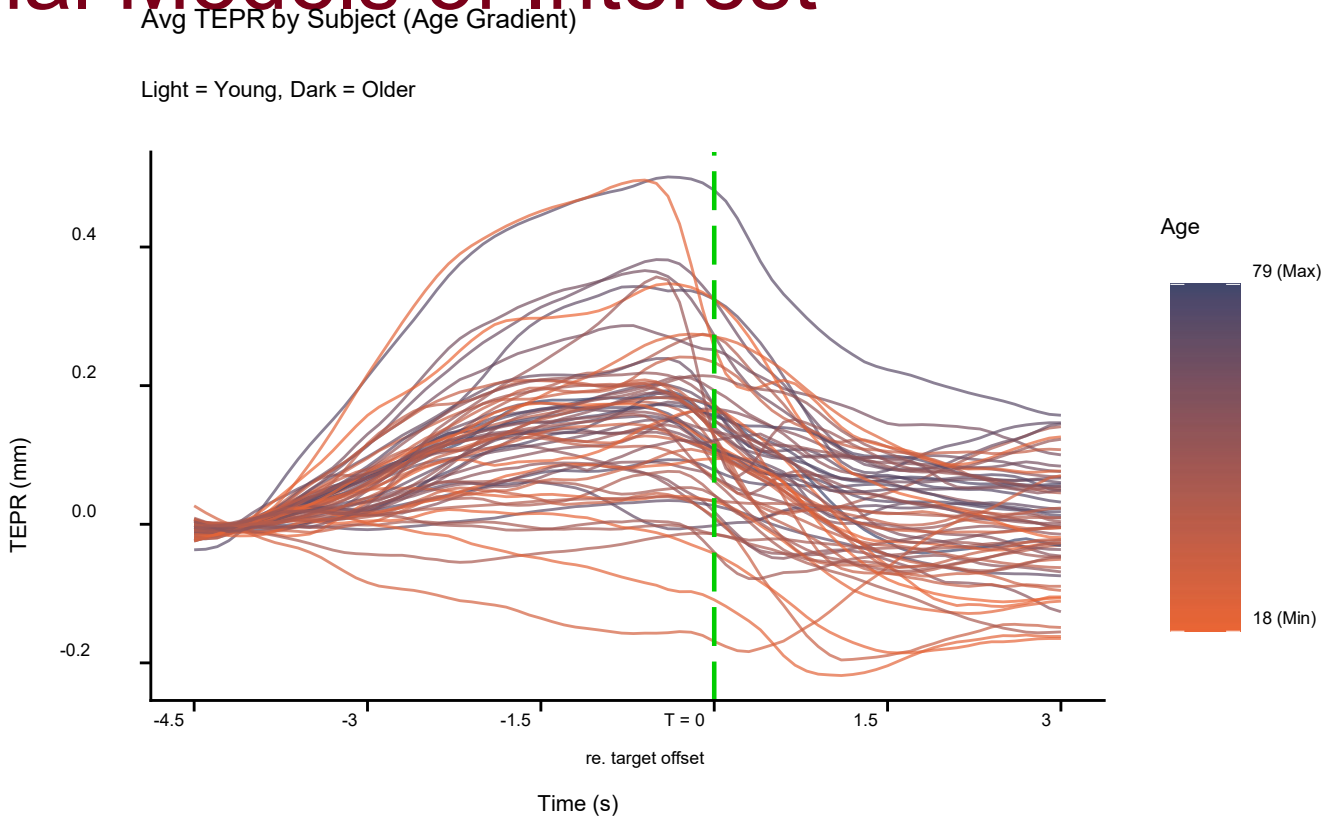


Summed Effects

Additional Models of Interest

(curiosity..)

- Younger / older split
- Identical formulas



Q’s for JM

- Units for raw data vs units for figs based on smooths
- How to explain significance of terms; we know it assesses “wiggliness”, however in the context of our data what does that mean? That the smooth is able to sufficiently account for the variance in the data?

Data

- Larger study is interested in EEG, self-report measures of LE, and more…
- Present analysis focused on pupillometry data
- N = 62, mean age = 47.6; SD = 17.65